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Problem Set for Tutorial 4

To be discussed on T4, October 25, 2019

1. Read carefully the following article and answer the following *two* questions.

15 July 2012

Zara good, Funeral bad: What companies are going to charge VAT (value added tax) in the final price?

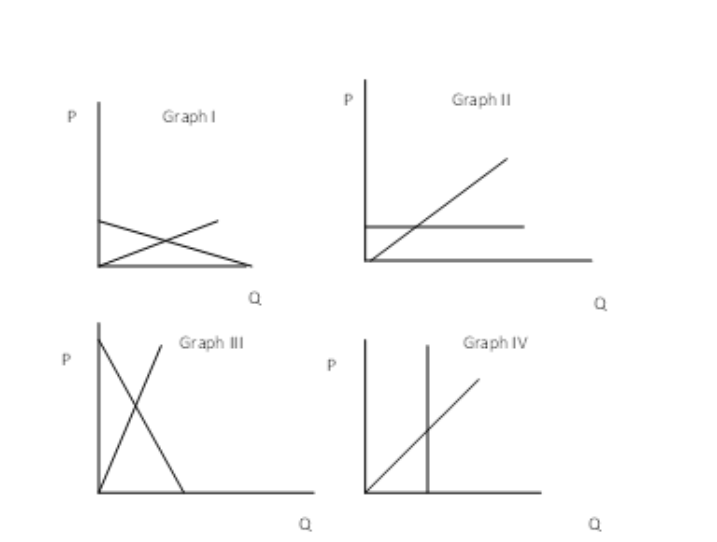
Written by Xavier Sala-i-Martin

VAT is fashionable. Since the European Union rose Spain's VAT, the whole country has given its opinion about the impact that this rise will have on the pockets of its citizens. Most analysts have done a simple calculation: if a product that, before tax, was worth 100 euros and with a 18% VAT was 118 euros, now with a VAT of 21% it will cost 121. That is, the consumer will have to pay 3 euros more to buy that product. The exercise is repeated for 15 different products and fear is put into the body of all listeners.

However, **this type of analysis has a slight drawback: It's wrong!** And it is wrong because not all companies "will pass" the tax increase in the final price that they charge consumers. In fact, **Zara**, has already been quick to announce that they would come out in defense of their clients and not increase prices so that the VAT increase would end up having an impact on lower profits for the company. The **funeral services**, on the other hand, explained that they would be "forced" to raise prices paid by the final consumer (in this case, the relatives of the final consumer).

But neither Zara or funeral homes tell the whole truth. It is not true that Zara will not rise prices (only) because it is defending consumers. Nor is true that funeral homes raise prices (only) because they are forced to do so. **Both respond to market conditions.**

Look at the following graphs and answer the following questions:



- (a) According to what the two firms say, which graph most accurately represents Zara? Which graph most accurately represents funeral homes?

Answer: *Graph II; Graph IV.*

- (b) The increase in VAT will be absorbed by the consumers in the market because the quotient $|\epsilon_p^{\text{demand}}|/|\epsilon_p^{\text{supply}}|$ is that that same quotient in the market.

Answer: *Funeral homes; lower; Zara*

2. In some states of the United States of America, legislation has changed, giving the now the same treatment to marijuana and alcohol. The consumption, production and exchange of marijuana with recreational purposes is now legal.

Suppose that initially the demand and supply of marijuana were the following

$$Q_D = 1000 - 2P$$

$$Q_P = -100 + 2P$$

With the new legislation the production costs of marijuana have decreased and the demand has also increased so that the equations of demand and supply after the approval of the new legislation are:

$$Q_D = 1200 - 2P$$

$$Q_S = 2P$$

With the previous information answer the following three questions:

- (a) If the objective of the state governments is that the same quantity of marijuana is consumed before and after the legislation comes into force, the per unit tax on marijuana needs to be \$, % of the tax will need to be paid by producers, and % of the tax will need to be paid by consumers.

Answer: *\$150; 50%; 50%*

- (b) After the drug legalization, which of the following quantities:

- $Q = 225$
- $Q = 300$

maximizes government revenues? The percentage of the tax that will be paid by producers is and the the percentage paid by consumers is .

Answer: *300; 50%; 50%*

- (c) Which of the following options minimizes the dead-weight loss associated with the imposition of a tax on marijuana?

- A tax of \$200 per unit.
- A tax of \$300 per unit.
- A tax of \$150 per unit.
- A tax of \$100 per unit.

Answer: *Establish a tax of \$100 per unit exchanged.*

3. The marginal cost of planting a tree is constant and is equal to $MC = 25$ dollars. There are 15 people living in a small town who have different preferences over the trees planted on the streets. Trees are a public good. Five people have high marginal benefits given by $MB_H = 100 - 0.5Q$ and the remaining ten people have low marginal benefits $MB_L = 50 - 0.25Q$, where Q is the number of trees planted on city streets.

- (a) Plot individual MB s and the MC on a diagram. What is the maximum number of trees that can be potentially privately provided in this town if people do not share costs? Do you think it is possible that no trees will be privately provided?

Solution. MC intersects MB_L at $Q = 100$ and intersects MB_H at $Q = 150$. If people do not share cost, then each person who decides to plant a tree has to incur in a private cost of \$25 per tree. Recall the idea behind individual choice whether to buy a unit or not: each consumer will be willing to pay for an additional tree as long as he gets marginal benefits from the tree at least as high as the cost. This means any person with low marginal cost will potentially buy 100 trees and each high marginal benefits person will be willing to buy up to 150 trees.

Notice the following. Since trees are a public good, once a tree is planted, everybody can enjoy the benefits of it. This means that the maximum number of trees that will be privately planted cannot exceed 150. If there is one person in the city who unwittingly plants 150 trees, that will be the end of the story. At $Q = 150$ $MB_L = 12.5$, clearly none of the consumers with low marginal benefits will be buying any more trees. How about the guys with high value of the trees? Suppose a person looks at the streets and sees 150 trees. If he buys one more tree, it will be the 151st tree with $MB_H = 150 - 0.5 \times 151 = 25.5$, which is lower than the cost of the tree—clearly this 151st tree will never be planted if it is left to **private decisions**. Thus, the conclusion is that we cannot tell who exactly will decide to plant how many trees, but the possible maximum number of trees privately provided is 150. The actual number of trees maybe smaller: even though it seems rational to buy up to 150 trees even without sharing the cost, smart consumers will free-ride and wait for someone else to provide the public good. ■

- (b) Find and plot on a diagram the social and marginal benefit curve. Calculate the socially optimal number of trees planted in town.

Solution. $MSB = 5 \times MB_H + 10 \times MB_L = 5(100 - 0.5Q) + 10(50 - 0.25Q) = 1000 - 5Q$. Efficiency requires that $MSC = MSB$, that is, $25 = 1000 - 5Q$, so $Q = 195$ trees. ■

- (c) How will your answer to parts (a) and (b) change if the number of people in town increases?

Solution. In part (a), if anything will change at all, it will be to the worse: the more people there are the more likely they are to expect somebody else to plant the trees. In part (b) the more people there are the higher will be the MSB of each tree and therefore the higher the optimal number of trees. ■